

The Depth of Traversal

Why a Closed Path Cannot Reverse the Order of Its Own Traversal, from Gödel Spacetime to the AdS/CFT Domain

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This paper consolidates three earlier working papers of the same programme: “The Depth of Traversal: Why a Closed Timelike Curve Does Not Reverse the Order of Experience”, “Traversal Depth Across the AdS/CFT Domain”, and “Geometric Lift of Negative Holographic Evolution”. It supersedes all three. The argument, the examples and the language are carried forward from the first of them, because the point of the argument is its simplicity.

The paper is in two halves separated by a printed line. Parts One to Three use standard relativity, published theorems and cited models, and stand or fall on that. Part Four is an extension in the author's own framework, Unified Mechanics, Corpus Edition 2026, Fifth Printing, and is offered as interpretation. Nothing above the line depends on anything below it.

A closed path is not automatically a reversed journey. A system may return to a labelled point while still carrying the completed traversal that brought it there. This paper develops that distinction from the observer outward: age, memory, entropy, physical state, the geometry of the loop itself, and finally the same structure as it appears in holographic reconstruction.

Abstract

Gödel-type spacetimes permit closed timelike curves: future-directed timelike paths that return to their starting spacetime event. The usual picture is a circle. This paper argues that the circle suppresses a physically decisive dimension, the depth created by traversal. An observer moving through $A \rightarrow B \rightarrow C$ does not encounter the original A again as an unexperienced occurrence. The return is A after B and C , and therefore belongs to a new traversal index. When that index is retained, the circle unfolds into a helix: the relational pattern repeats, but the occurrence does not. Recent quantum treatments make the same pressure visible by requiring the traveller's state, memory and entropy to return to their initial values. Such recurrence may secure mathematical consistency, but it does so by imposing a reset or reversal on the very history that would make the trip a meaningful return to one's own past.

The depth argued for here cannot be supplied by topology. Gödel's manifold is simply connected, so its closed timelike curves cannot be opened by passing to a covering space. The index must be carried by the traversal itself. The second half of the paper shows that this same structure appears, in an entirely different setting, in holographic Hamilton–Jacobi evolution. Boundary-to-bulk reconstruction moves opposite to the conventionally outward radial vector field, but a negative geometric direction is not negative completed evolution. Once the boundary is marked as the origin, the sign-definite decrease can be represented exactly by a positive accumulated variable, and more

strongly by adjoining a monotone history coordinate so that the negative base flow is retained exactly while the completed operation rises. The holographic trajectory then has the structure of the spring: its projection may repeat a radial label or a Jacobi-space configuration, while its origin-marked traversal remains ordered.

The central claim is interpretive and physical rather than algebraic: geometric closure does not by itself establish reversal of elapsed evolution.

Keywords: closed timelike curves; Gödel spacetime; traversal depth; AdS/CFT; Hamilton–Jacobi flow; holographic renormalization; geometric lift; positive evolution.

PART I • THE OBSERVER

1. The Problem Begins with the Observer

Gödel supposed a universe in which the global structure of spacetime permits closed timelike curves. This is usually described as a route through spacetime that eventually returns an observer to the same event. But the observer is not merely a location on the route. The observer is a physical system undergoing the route.

The first question is deliberately ordinary: what happens to the observer’s age? “Age” here means exactly what it normally means, the time the observer has actually lived through. The observer’s body changes, their clocks advance, their memory changes, and their physical state accumulates the journey. These changes are not external annotations placed on the curve. They are part of what it means for an observer to traverse it.

This creates the basic tension. The geometry says that the route closes. The observer says that the route happened. If the traveller reaches the labelled starting point only after completing the journey, then the returned occurrence already contains a history that the original occurrence did not.

2. Geometry and Physics Must Make Sense Together

The argument is not that Gödel’s metric fails as a solution of Einstein’s field equations. The argument is that a geometrically closed path and a physically identical recurrence are not the same claim. Geometry may identify endpoints. Physics must still explain the state of the system that arrives there.

Let an observer follow a timelike worldline

$$\gamma : [0, T] \rightarrow M, \quad \text{with} \quad \gamma(T) = \gamma(0).$$

The equality states geometric closure: the endpoint and starting point are the same event in the spacetime manifold. Yet the observer’s proper time along the path is

$$\Delta\tau = \int_0^T \sqrt{(-g_{uv} \dot{\gamma}^u \dot{\gamma}^v)} d\lambda > 0.$$

The observer therefore undergoes a nonzero duration while travelling from the start back to the start. The two statements can coexist mathematically, but they do not mean the same thing:

$$\gamma(T) = \gamma(0) \text{ does not mean } \Delta\tau = 0.$$

The route can close while the traversal remains completed. The physical question is what becomes of that completed traversal when the endpoint is identified with the beginning.

3. Existing Quantum Models Reveal the Same Pressure

David Deutsch modelled a quantum system interacting with a closed timelike curve by requiring the state on the curve to be a fixed point of the total interaction. In a standard notation,

$$\rho_{CTC} = Tr_{ext}[U(\rho_{in} \otimes \rho_{CTC})U^\dagger].$$

This condition does not describe an unrestricted observer simply moving backward through their own history. It requires the loop state to be self-consistent after the full interaction. The state is permitted only if the end of the loop reproduces the state required at its beginning.

A more recent analysis by Lorenzo Gavassino places an isolated spaceship and observer on a closed timelike curve in an axially symmetric setting. The internal quantum system is constrained so that it returns to its initial state after the round trip. Memory must be erased by the end of the journey, and any entropy increase must eventually be reversed so that the loop closes consistently.

This is a brilliant result because it exposes the issue rather than removing it. The model can close only by making the traveller's accumulated physical history unwind. That immediately raises the human question: if the observer's memory, ageing and internal change are all reversed before closure, in what meaningful sense did that same observer return carrying a journey into their past?

The mathematics describes a self-consistent recurrence. It does not automatically establish the ordinary idea of a traveller who remembers leaving, travels around the loop, and then arrives in their own earlier world as the same continuously accumulated person.

4. The Circle Hides the Order

The usual representation of a closed timelike curve is a circle:

$$A \rightarrow B \rightarrow C \rightarrow A.$$

But an observer cannot experience this sequence as a simultaneous circle. The observer experiences it in order: A, then B, then C, then the return toward A. Once C has occurred, the next A cannot be the unexperienced original A from the observer's point of view. It is A after C.

The moment C recognises A as the start of a new loop, a boundary has already recorded the previous set. The completed traversal distinguishes the returned A from the first A. The labels may repeat, but the occurrences do not.

$$A_0 \rightarrow B_0 \rightarrow C_0 \rightarrow A_1 \rightarrow B_1 \rightarrow C_1 \rightarrow A_2 \rightarrow \dots$$

This indexed sequence is not an extra story imposed on the traveller. It is the minimal record of the order the traveller must actually undergo. If the indices are erased, the geometry appears circular. If they are retained, the route has depth.

5. Giving B Depth

In the flat circle, B is treated only as a midpoint or relation between A and C. But once traversal is preserved, B does more. B is the passage by which one occurrence of the pattern becomes the next occurrence. It is therefore the depth-bearing relation.

The first traversal is

$$A_0 \rightarrow B_0 \rightarrow C_0.$$

The return does not collapse this sequence. It completes it and begins another:

$$C_0 \rightarrow A_1 \rightarrow B_1 \rightarrow C_1.$$

B now behaves like a stair rather than a flat point. Each B carries the system across the interval separating one completed level from the next. The repeated A–B–C pattern remains visible around the width of the structure, while the traversal index increases through its depth.

Repetition preserves the shape of the loop; traversal gives the loop depth.

Equivalently: the circle only appears closed because the depth of traversal has been removed. Once B retains the passage already completed, each return to A occurs on a new level. The circle opens into a spring or helix.

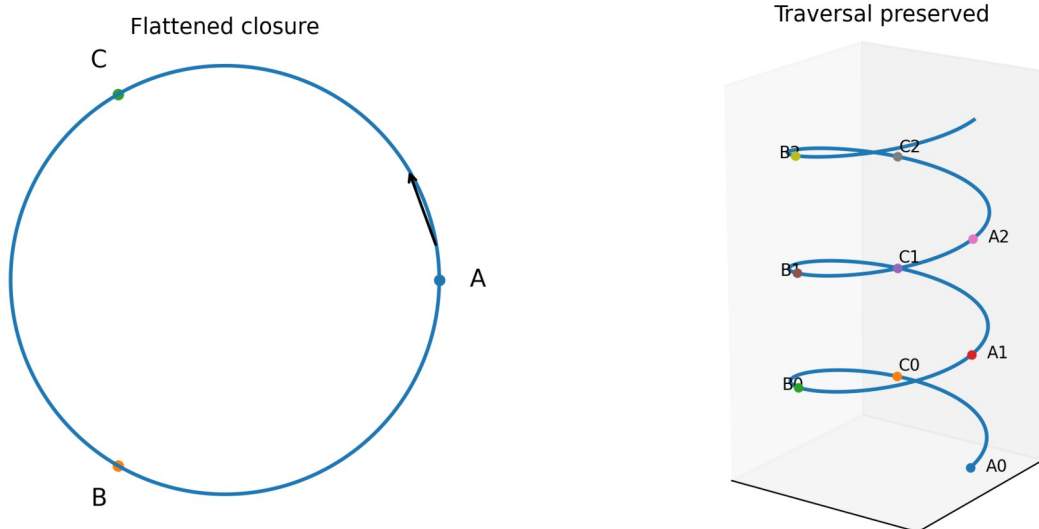


Figure 1. The same A–B–C relation shown first with traversal suppressed and then with traversal retained.

6. A Compact Mathematical Unfolding

The distinction can be expressed using a circle and its unwrapped traversal. Let the geometric loop be parameterised by an angle

$$\theta \in S^1, \quad \theta \equiv \theta + 2\pi.$$

On the circle, θ and $\theta + 2\pi$ are identified. That identification records position on the loop but suppresses how many times the loop has been traversed. Introduce an unwrapped parameter $s \in \mathbb{R}$ and the projection

$$p(s) = e^{is} \in S^1.$$

Then

$$p(s) = p(s + 2\pi n), \quad n \in \mathbb{Z}, \quad \text{even though} \quad s \neq s + 2\pi n.$$

This is the mathematical version of the argument. The same geometric point can be reached at distinct positions in the ordered traversal. The projection to the circle identifies them; the unwrapped parameter distinguishes them.

The A–B–C labels can be assigned by phase. For example,

$$A_n = 2\pi n, \quad B_n = 2\pi n + 2\pi/3, \quad C_n = 2\pi n + 4\pi/3.$$

The experienced order is therefore

$$A_n < B_n < C_n < A_{n+1}.$$

No step in this order is $C \rightarrow B \rightarrow A$. The traveller moves continually forward in the unwrapped parameter even though the projection repeats positions on the circle.

7. The Helix as the Physical Picture

A helix preserves both facts at once: cyclic repetition and forward traversal. One simple embedding is

$$H(s) = (R \cos s, R \sin s, hs), \quad h > 0.$$

The first two coordinates reproduce the circle. The third coordinate records depth. After one complete turn,

$$H(s + 2\pi) = (R \cos s, R \sin s, hs + 2\pi h),$$

so the horizontal position repeats while the full occurrence does not. The helix does not deny cyclic structure. It denies that cyclic structure alone is sufficient to erase traversal.

In this picture, the ordinary circular closed timelike curve is a projection of a deeper ordered structure:

$$\pi(H(s)) = (R \cos s, R \sin s).$$

Flattening the helix removes the coordinate that records the completed journey. The result looks closed because the information that distinguishes one turn from the next has been discarded.

8. State Evolution and the Non-Reversal of Occurrence

Let $X(s)$ denote the complete physical state of the traveller: body, clocks, memories, entropy and every other internal degree of freedom. Ordinary evolution along the traveller's proper time may be written abstractly as

$$dX/d\tau = F(X).$$

A genuinely identical closure requires

$$X(T) = X(0).$$

That equality is stronger than returning to the same geometric event. It requires the total physical evolution over one loop to be periodic. In operator form,

$$X(T) = \Phi_T(X(0)), \quad \text{with} \quad \Phi_T(X(0)) = X(0).$$

The initial state must therefore be a fixed point, or belong to a periodic orbit, of the complete evolution. This is precisely why closed-timelike-curve models require consistency conditions, memory erasure, entropy reversal or specially quantised dynamics. These mechanisms do not show that elapsed history becomes logically equivalent to no history. They enforce recurrence by constraining which histories are allowed.

The distinction is simple:

$$\text{same geometric location} \neq \text{same occurrence} \neq \text{same accumulated state}.$$

9. Why the Depth Cannot Be Supplied by Topology

There is a natural objection at this point, and answering it shows what the argument actually requires. The objection is that unwrapping a loop into a helix is a familiar operation. Where a closed timelike curve arises from an identification of points, one may pass to the universal covering space and the loop opens by construction. Anti-de Sitter space is the standard example. Its timelike circle is a consequence of an identification, the covering space undoes that identification, and the closed curve is gone. On this reading the helix is not an argument but a change of chart.

Gödel's universe does not permit that move. The manifold is topologically $\mathbb{R}^4$ and simply connected. Its closed timelike curves are not produced by identifying distinct points. They are produced by the tilting of the light cones at sufficient distance from the rotation axis, so that a future-directed path can return to its own starting event with no identification made anywhere. The covering space of Gödel spacetime is Gödel spacetime. There is no sheet to climb to.

This is the decisive point for what follows. If the loop is to be opened, the depth cannot come from the manifold, because the manifold has none to give. It must come from the traversal. The index in

$$A_0 \rightarrow B_0 \rightarrow C_0 \rightarrow A_1 \rightarrow \dots$$

is therefore not a sheet number in a covering space. It is a record of what has been undergone by the system moving along the curve. B is the depth-bearing relation because B is what the observer performs, not because B lies on a different sheet.

Read this way, Gödel forces the question that the covering-space construction normally allows one to avoid. When there is no topological room to store the completed traversal, where does it go? The answer offered here is that it is stored in the state, and that the consistency conditions of Section 3 are the price a model pays for insisting that it not be stored anywhere.

PART II · THE SAME STRUCTURE IN THE ADS/CFT DOMAIN

10. The Minimal Claim, Carried Forward

The claim carried into the second half of this paper is the weak one, stated before any specialised machinery is introduced. If a system passes through an ordered sequence and returns to a label already occupied, the completed passage remains part of the resulting state.

You went around once. Even if you are back at A, this is now the next occurrence of A.

In symbols,

$$A_0 \rightarrow B_0 \rightarrow C_0 \rightarrow A_1, \quad \text{with} \quad A_1 \neq A_0 \text{ as an occurrence.}$$

The subscript records no additional physical assumption. It retains only the order already required to complete the traversal. The endpoint may share the same projected label while remaining distinct as an occurrence. Nothing in this statement is specific to spacetime, and that is why it travels.

11. Projection and Lift

Let $q(s)$ denote the state seen in the ordinary geometric or configuration-space description. Introduce a history variable $E(s)$ that records completed traversal, with

$$dE/ds = \kappa(s), \quad \kappa(s) > 0.$$

The complete state is then not $q(s)$ alone but the lifted state

$$\tilde{Q}(s) = (q(s), E(s)).$$

The projection forgets traversal depth:

$$\pi(\tilde{Q}(s)) = q(s).$$

A projected recurrence can therefore satisfy $q(s_1) = q(s_0)$ while the complete states remain different:

$$\tilde{Q}(s_1) \neq \tilde{Q}(s_0), \quad \text{because} \quad E(s_1) - E(s_0) = \int_{s_0}^{s_1} \kappa(u) du > 0.$$

This is the general circle-to-spring statement. A base description may close or recur. A strictly increasing history coordinate cannot close merely because the projection does.

12. Marking the Boundary as the Origin

In the Hamilton–Jacobi treatment of AdS/CFT, the bulk radial coordinate acts as an evolution parameter. Boundary data are supplied on a radial slice and used to reconstruct the bulk. The usual first-order Hamilton–Jacobi vector field is conventionally written in the outward radial direction.

Let q denote the collection of induced fields on a radial slice: metric data, scalar fields and any other variables retained in the truncation. Write the conventional outward flow as

$$dq/dr = V_{HJ}(q).$$

Now mark the boundary as the operational origin. If $r = r_b$ at the boundary, define an inward parameter

$$s = r_b - r, \quad \text{so that} \quad s = 0 \text{ at the boundary and } ds = -dr.$$

The same geometric trajectory, performed from boundary to bulk, obeys

$$dq/ds = -V_{HJ}(q).$$

This minus sign does not say that the reconstruction un-happens. It says that the operation proceeds opposite to the conventionally outward radial orientation. The boundary is A, the radial reconstruction is B, and the reached bulk configuration is C. B gives the trajectory depth because it is the operation that carries the boundary data into the bulk.

13. From Negative Direction to Positive Completed Evolution

Suppose a Hamilton–Jacobi quantity $W(q(s))$ decreases along the boundary-originated branch:

$$dW/ds = -K(q(s)), \quad K(q) \geq 0.$$

Integrating from the marked boundary origin gives

$$W(s) - W(0) = -\int_0^s K(q(u)) du.$$

The negative integral records direction in the Hamilton–Jacobi landscape. It is not a negative amount of evolution. Define the positive completed evolution

$$E(s) := \int_0^s K(q(u)) du = W(0) - W(s) \geq 0, \quad \text{so that} \quad dE/ds = K(q(s)) \geq 0.$$

Nothing has been discarded. The minus sign has been moved from the amount completed into the orientation of the base trajectory. The exact same branch can therefore be written in two complementary ways:

$$\text{geometric direction: } dW/ds = -K. \quad \text{completed evolution: } dE/ds = +K.$$

This is the holographic version of opening the circle into a spring. The radial coordinate may run inward, or a Jacobi-space quantity may decrease, while the traversal performed from the marked origin accumulates positively.

14. Keeping the Minus Sign: the Geometric Lift

The rectification of the previous section is exact, but it can look as though a negative equation has been replaced by a positive one. It has not, and the stronger formulation makes that visible by keeping both quantities together in one geometry rather than trading one for the other.

Let Q be the original Hamilton–Jacobi configuration space. Introduce the lifted space

$$\tilde{Q} = Q \times \mathbb{R}_+,$$

where the added coordinate H records traversal depth from the marked origin, with $H(0) = 0$. Choose any nonnegative rate that faithfully measures completed motion. The most direct geometric choice is the speed of the base trajectory with respect to a positive metric G on the relevant configuration-space directions:

$$dH/d\tau = \|dq/d\tau\|_G = \sqrt{[G(dq/d\tau, dq/d\tau)]} \geq 0.$$

For a base flow $dq/d\tau = -F(q)$, the lifted flow is

$$d\tilde{q}/d\tau = (-F(q), \|F(q)\|_G).$$

The minus sign has not been dropped. It remains the first component of the lifted vector. Positivity appears as the second component, which measures the amount of operation completed along that negative base direction. Written as an operation on vector fields,

$$L_+[V](q, H) = (V(q), v(q)), \quad v(q) \geq 0, \quad \text{so that} \quad L_+[-F] = (-F, v).$$

This is the precise form of raising the operator. The base operator remains $-F$. The lift embeds it into a higher-dimensional tangent space where the final component lies in the positive history direction. The lifted operator therefore respects positive operation without pretending that its radial component is positive.

The two facts that seemed to be in conflict are now two components of one flow:

$$\text{base displacement: } dq/d\tau = -F(q). \quad \text{completed operation: } dH/d\tau \geq 0.$$

The first states where the system moves in the original configuration space. The second states that an operation initiated at an origin has accumulated rather than un-happened. The apparent conflict was produced only by asking one coordinate to represent both displacement and completion.

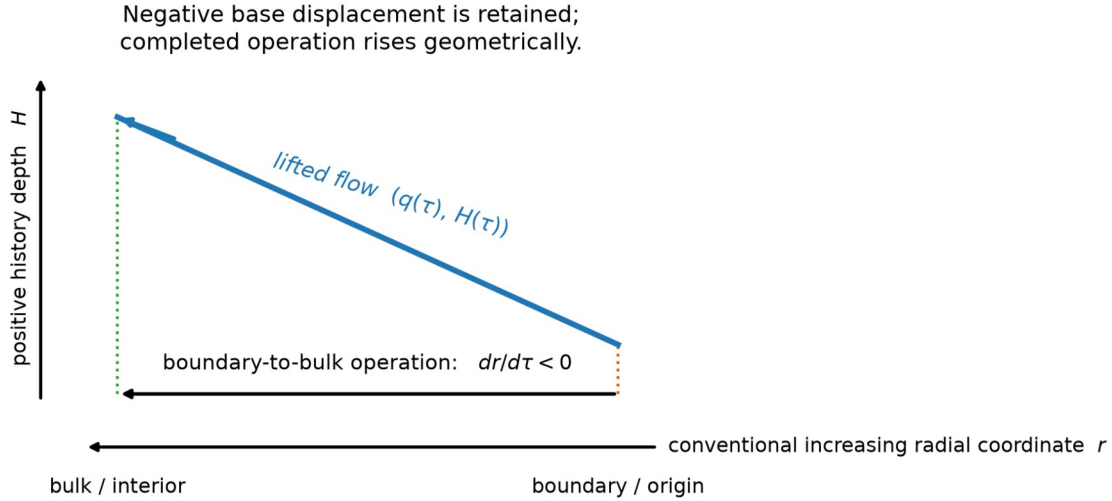


Figure 2. The radial coordinate decreases while the lifted history coordinate increases.

15. Choosing the Rate: Beta Functions and the Depth of B

The lift is structural. The choice of v determines what physical or operational quantity the added depth records. Several choices are available on a sign-definite holographic branch.

History rate	Meaning
$v = \ F\ G$	Configuration-space arc length; the most direct geometric traversal depth.
$v = -dW/d\tau$	Positive decrease of the Hamilton–Jacobi functional where $dW/d\tau \leq 0$.
$v = G_{a\beta} \beta^a \beta^b$	Positive holographic renormalization-group speed when the scalar target-space metric is positive definite.
$v = dL/d\tau $	Total variation of a local action density, useful when a signed branch changes orientation.

The third of these gives B its most explicit content. In a standard scalar holographic flow, with a positive-definite scalar target-space metric $G_{a\beta}$, the positive rate can be expressed through the norm of the boundary beta function,

$$K = G_{a\beta} \beta^a \beta^b \geq 0, \quad \text{hence} \quad E(s) = \int_0^s G_{a\beta} \beta^a \beta^b du.$$

This gives B a precise mathematical depth. The radial passage is not an empty connector between two endpoint descriptions. Its depth is the accumulated norm of the flow performed between them. At a fixed point, $\beta^a = 0$ and the coupling-space contribution does not run. Away from a fixed point, every nonzero beta-function norm contributes positively to completed evolution.

The lift adds no bulk field and changes no field equation. It augments the representation of the solution by retaining an accumulated readout of the operation.

16. A Note on the Name

The integrals used above are labelled Arrow of Time Operation Integrals, and the name should be read modestly. An ordinary product forgets the order in which its factors were performed. Writing the operators with a direction attached simply requires the first operation of a set to be respected as first, so that a product which produces a rate, and then a rate of that rate, carries its own order rather than having the order supplied afterwards by the reader.

That is all the name records. It is a bookkeeping convention for keeping order inside an operation, which is the same service the traversal index performs for the loop. Nothing in the argument of this paper depends on the convention; it is a notation for the thing already argued for, not an additional claim.

17. Finite Domains and Dimensional Incompleteness

Suppose the radial calculation is performed on a finite interval $r \in [r_{IR}, r_b]$. The negative derivative $dr/d\tau = -1$ exists throughout the interval and therefore cannot be attributed to leaving it. The deeper limitation is that the finite base domain records radial position but not traversal depth:

$$D = \{q(r) : r_{IR} \leq r \leq r_b\}, \quad \tilde{D} = \{(q(r), H) : r_{IR} \leq r \leq r_b, H \geq 0\}.$$

A trajectory can stay entirely inside D in its first component while advancing into values of H that D does not represent. In this precise sense the completed evolution lies outside the dimensional completeness of the base description, not outside its radial coordinate bounds. This is the same observation made in Section 9 about Gödel, stated for a domain rather than a manifold: the description can be complete in its own coordinates and still have nowhere to put the traversal.

18. Recurrence Becomes Vertical Separation

The lift also resolves the geometric ambiguity that appears when a base trajectory retraces or recurs. If

$$q(\tau_2) = q(\tau_1), \quad \tau_2 > \tau_1,$$

but nonzero traversal occurred between them, then $H(\tau_2) > H(\tau_1)$ and therefore

$$\tilde{q}(\tau_2) \neq \tilde{q}(\tau_1).$$

The base projection may form a loop, repeat a slice, or collapse two endpoints into one label. In the lifted space those same points are vertically separated by completed history. A circle in the projection becomes a helix when the traversal coordinate is retained. The lifted holographic state is

$$\tilde{Q}(s) = (q(s), E(s)),$$

and in the notation of Part I,

$$A_0 \rightarrow B_0 \rightarrow C_0 \rightarrow A_1 \quad \text{becomes} \quad (q_0, E_0) \rightarrow \text{radial reconstruction} \rightarrow (q_1, E_1), \quad E_1 > E_0.$$

Even when q_1 is identified with q_0 by a projection, a symmetry or a recurrence, the lifted state retains the completed operation. The holographic flow is helical in the same structural sense: the configuration may repeat across the width, while positive evolution advances through the depth.

PART III · THE COMMON PRINCIPLE

19. One Structure, Two Domains

Gödel-type closure and holographic radial reversal initially look like different problems. One concerns a closed timelike curve; the other concerns reconstruction across the AdS/CFT domain. Their logical structure is nevertheless the same: an endpoint or projected description can suppress the operation that produced it.

Structure	Projection says	Traversal-preserving lift says
Circle / Gödel path	A returns to A	A_0 reaches A_1 after B and C
AdS/CFT radial flow	The same trajectory is reparameterized	Boundary origin fixes the performed direction
Negative Hamilton–Jacobi change	W decreases	$E = W(0) - W(s)$ increases
Configuration recurrence	$q(s_1) = q(s_0)$	$(q, E)(s_1) \neq (q, E)(s_0)$ when E increased

The common rule is:

Projection may repeat an endpoint. Traversal preserves the operation.

Or in the language of the spring:

Repetition preserves the shape; positive evolution gives it depth.

20. A Compact Theorem

Theorem 1 (Positive traversal lift). Let $q : [0, S] \rightarrow X$ be a trajectory in a base state space, and let $K : X \rightarrow [0, \infty)$ be continuous. Define

$$E(s) = \int_0^s K(q(u))du, \quad \tilde{Q}(s) = (q(s), E(s)).$$

If $q(s_1) = q(s_0)$ for $s_1 > s_0$ and $K(q(u))$ is positive on a subset of $[s_0, s_1]$ of nonzero measure, then $\tilde{Q}(s_1) \neq \tilde{Q}(s_0)$.

Proof. Since $K \geq 0$ and is positive over a nonzero part of the interval, $E(s_1) - E(s_0) = \int_{s_0}^{s_1} K(q(u))du > 0$. Thus the history components differ even though the projected base states agree. Therefore the lifted states are unequal. ■

The same statement in vector-field form records that the base sign is untouched.

Theorem 2 (Sign-preserving form). Let q satisfy $dq/d\tau = V(q)$, and let $v(q)$ be continuous, nonnegative, and strictly positive wherever $V(q) \neq 0$. Define $H(\tau) = \int_0^\tau v(q(s))ds$ and $\tilde{q}(\tau) = (q(\tau), H(\tau))$. Then $d\tilde{q}/d\tau = (V(q), v(q))$ with $dH/d\tau \geq 0$. If the base flow is negative relative to a chosen orientation, $V = -F$, that sign remains unchanged in Q , and every nontrivial segment nevertheless rises strictly in the H direction.

For the circle, E may be turn number, elapsed proper time, or any strictly increasing traversal record. For holographic Hamilton–Jacobi evolution, E is the positive accumulated quantity obtained from the sign-definite branch. The theorem is the same in both domains.

21. What the Argument Does and Does Not Claim

This argument does not claim that the Gödel metric is algebraically inconsistent. Nor does it claim that periodic physical systems are impossible. A pendulum, orbit or quantum state may recur.

The claim is narrower and more direct: a closed timelike curve cannot be interpreted as a reversal of the observer’s experienced order merely because the spacetime path closes. The observer still undergoes A before B , B before C and C before the returned A . To identify that returned A with the original occurrence, the theory must additionally erase, reverse or exactly recur the entire physical state that distinguishes them.

Once this extra requirement is made explicit, the usual time-travel picture changes. What has been demonstrated is not unrestricted travel backward through an already lived sequence. It is a globally constrained periodic process whose admissible states are selected to make the loop self-consistent.

On the holographic side the corresponding restraint applies. The construction does not claim that every negative derivative is secretly positive. It does not reverse a radial coordinate convention, discard a minus sign, alter the Hamilton–Jacobi constraint, or introduce an independent physical time into Euclidean AdS/CFT. It establishes a narrower statement: signed displacement and positive completion are compatible components of one lifted operation. The history coordinate is derived

from the path and the origin, and its interpretation depends on the selected rate v and on the regime in which positivity holds. If the relevant metric is indefinite, the flow changes branch, or corrections spoil sign-definiteness, the rate must be reconsidered or defined through branchwise total variation.

It is worth saying plainly what this paper declines to do. Once the traveller's memory, entropy and internal state are made the subject, it is easy to keep going: to ask what the erasure costs, where the heat is carried, which microstates are admissible. Those threads exist and they can be pulled. They are not pulled here, because the argument does not need them and they obscure it. The point is settled at the level of order, before the machine is opened.

22. Conclusion

The answer is not going to be located in an isolated quantum state already buried inside a model designed to defeat paradox. It begins one level earlier, where geometry and physics are required to make sense together.

A circle records recurrence of position. An observer records traversal. The moment the observer passes from A through B to C, the return to A has acquired depth. The repeated label A does not make the occurrence identical; it marks the start of a new turn. B becomes the stair between turns, and the flat loop becomes a helix. In Gödel's universe that depth cannot be supplied by the manifold, because there is no covering space to supply it. It is supplied by the traversal.

The holographic case is the same structure met in a different setting. Moving opposite to the outward radial Hamilton–Jacobi field does not make the boundary-to-bulk operation run backward. It gives the base trajectory a negative orientation while the completed evolution accumulates positively:

$$-F(q) \rightarrow (-F(q), v(q)), \quad v(q) \geq 0.$$

The complete flow, in both domains, is the same:

mark the origin $\rightarrow$ preserve the direction $\rightarrow$ perform the traversal $\rightarrow$ accumulate positive depth $\rightarrow$ distinguish the returned projection from the completed history.

The final conclusion is therefore concise:

A closed path may repeat its geometry, but it cannot reverse the order of its own traversal. To make the end identical to the beginning, a theory must remove or reverse the physical history that made the journey occur. That may produce a consistent loop. It does not, by itself, produce meaningful travel into one's own already-lived past.

ABOVE THIS LINE: STANDARD RELATIVITY, PUBLISHED THEOREMS AND CITED MODELS
BELOW THIS LINE: THE AUTHOR'S OWN FRAMEWORK, OFFERED AS INTERPRETATION

PART IV · EXTENSION · THE UNIFIED MECHANICS READING

23. The Line, and What Is Below It

Everything to this point uses standard relativity, published theorems and cited models. Gödel's metric is Gödel's. The consistency conditions are Deutsch's and Gavassino's. The lift is elementary. A reader who rejects everything that follows keeps all of it, because nothing above the line depends on anything below it.

What follows is the author's own framework, Unified Mechanics, Corpus Edition 2026, Fifth Printing, and is offered as interpretation. The Corpus grades are used throughout: PROVED follows from stated definitions or assumptions; CONDITIONAL follows once a named condition is granted, with the condition written into the statement; PROPOSED is a physical reading offered for calculation and falsification; OPEN names what can be formulated but not yet settled.

Part One left three joints weaker than the rest, and this part addresses them. It never said what fixes the traversal index rather than leaving it a matter of bookkeeping. It said the depth is stored in the state without saying in what. And it argued that memory erasure makes the returned observer a different observer, which read as a rhetorical question rather than a claim. Each of those has a theorem behind it, and two of the theorems are not the author's.

24. Causal Order Determines the Geometry

The argument of Part One has an obvious line of attack, and it should be met rather than avoided. A critic grants that the observer undergoes A before B before C, and replies that this is a fact about the observer and not about the spacetime. The geometry closes. Experience is a further story told about the geometry. Privileging the story is not physics.

The reply is a theorem, and it is not the author's.

Alexandrov and Zeeman: any bijection of a flat continuum of more than two dimensions that preserves causal order is a Lorentz transformation composed with a translation and a dilation.

Read what that says. The causal order determines the metric, up to the dilation that is a choice of unit. Not constrains it. Determines it. Malament extends the same conclusion to curved spacetime: for spacetimes that are past and future distinguishing, a causal isomorphism is a conformal isometry.

So the order is not a soft layer resting on a hard geometric substrate. The geometry is what one recovers from the order, and it is recoverable from nothing weaker. The objection therefore reverses. A model that erases the traversed order in order to close the loop is not preserving geometry while discarding a story about experience. It is discarding the datum from which the geometry was reconstructed, and keeping a coordinate description that the discarded datum was supposed to license.

PROVED and cited, not the author's. Alexandrov and Zeeman for the flat case, carried in the Corpus Register as T18, second witness for the Lorentzian signature. Malament for the curved case, brought in here rather than taken from the Corpus.

25. Gödel Fails the Hypothesis, and Fails It for the Reason Part One Names

Malament's theorem carries a hypothesis: the spacetime must be past and future distinguishing, meaning that distinct points have distinct causal pasts and distinct causal futures.

Gödel's universe fails that hypothesis, and fails it comprehensively. A spacetime with closed timelike curves through a region cannot distinguish the points on such a curve by their causal past or future, because each lies in the past and the future of the others. The causal order has been allowed to close, and a closed order is not an order.

This is worth stating plainly because it is the same observation Part One makes, arriving from the direction of an established theorem rather than from the observer.

In Gödel, the causal order does not determine the geometry. It fails to determine it for exactly the reason Section 4 gives: the order has been permitted to return to itself.

So the situation is not that geometry is secure and experience is soft. It is that the closure has damaged the causal order badly enough that the standard reconstruction of geometry from order no longer applies. Something has to be restored before the reconstruction is available again.

PROVED. That Gödel spacetime is not causally distinguishing is standard, and that Malament's hypothesis therefore fails is immediate. The reading of this as support for Part One is PROPOSED.

26. The Lift Restores the Distinguishing Condition

The lift of Part Two adjoins a monotone history coordinate. Its effect on the distinguishing condition is direct.

Let E increase strictly along every causal curve, which is what $dE/ds = K > 0$ says. Two points of the base with the same label, $q(s_1) = q(s_0)$, have $E(s_1) > E(s_0)$ and are therefore distinct in the lifted space.

No lifted point lies in its own causal past, because that would require E to return to a value it has left. The lifted space is past and future distinguishing by construction.

The lift is not decoration on the geometry. It is the operation that restores the hypothesis under which causal order determines geometry at all.

That is a stronger statement than Part Two made for itself. Part Two argued that the lifted states are unequal, which is true and, as Section 21 concedes, true of every flow. The point here is narrower and sharper: the lift is exactly what a spacetime with closed timelike curves is missing, measured against the condition an established theorem requires.

PROVED given that E is strictly increasing along causal curves, which holds wherever $K > 0$. Where K vanishes on a set of positive measure the conclusion fails on that set, and the framework says nothing there.

27. The Index Is Single, by Theorem

Part One assigns an index and the index does the work: A_0, A_1, A_2 . The second line of attack asks what fixes it. Indices are cheap. If the traversal record admits many labellings, then choosing one and calling the result depth is bookkeeping.

The Corpus proves it is single.

T15. The local clock mode is single. PROVED. The record accumulates along one monotone order; two independent orders of accumulation would be two records and therefore two systems.

That is the statement Part One needed and did not have. There is not a family of admissible traversal indices with one selected by convention. There is one, and the reason there is one is that a system with two would not be one system. The monotonicity is not imposed either; it is part of what the theorem asserts.

PROVED in the Corpus Register, from Movement Eleven and the arrow of Movement Thirteen. Its application to the traversal index of Part One is PROPOSED.

28. Closure Cannot Reach the Order Mode

Definition 18 of the Register places the geometry projectors mutually orthogonal: P_X onto external distinction, P_T onto local causal order, P_I onto system closure, summing to the geometry block. With the clock and closure modes each single by T15 and T16, the block splits three, one and one.

Orthogonality is the whole of Section 4's argument, stated as a definition rather than drawn as a picture. A closed path closes in the spatial projection. P_T is orthogonal to P_X , so closure in P_X has no component along the order mode and cannot act on it. The flattening that makes the loop look

closed is precisely the projection that drops P_T , and it drops it because it is orthogonal, not because it is small.

D18 carries a further clause that Part Two needed and did not have:

The physical projector is P_X plus P_T ; P_I is retained as closure data and excluded from traversable directions.

Retained, and not traversable. That is the lift stated in the Corpus's own terms, written before the lift was. A coordinate that records what has closed, carried in the description, and not a direction along which anything moves. Section 14 of Part Two describes the same object and had to argue for it; here it is a definition.

The rank also settles a question Part Two left open. The lift adjoins one monotone coordinate, and the choice of one was never defended, since any number would have produced a nonclosure theorem. The causal order projector has rank one by T15. The lift adjoins exactly one coordinate because there is exactly one order of accumulation.

D18 is a definition of the Register and T15 and T16 are PROVED. The identification of the lift's history coordinate with P_I , and of its rank with that of P_T , is PROPOSED.

29. Erasure Is Overwrite, and Overwrite Is Forbidden

Section 3 of Part One observes that Gavassino's loop closes only by erasing the traveller's memory and reversing the entropy, and asks in what meaningful sense the same observer returned. That was posed as a question. The Corpus answers it as a theorem.

T26. Retention over replacement. PROVED. A persistent self-description must join each new resolution to the record rather than overwrite it, or its later states are not records of the same system.

The consistency condition requires overwrite. T26 says a system that overwrites does not have later states that are records of the same system. So the two requirements are not in tension over a matter of interpretation. They are incompatible, and the incompatibility is what Part One was pointing at without the means to say so.

The Register also names the geometric form of the same prohibition:

T28. No recollapse while the record stands. Contraction of the causal domain toward coincidence would press resolved distinctions back into unread simultaneity, which is overwrite by geometry and is forbidden by T26.

Overwrite by geometry is exactly the case at issue. A closed timelike curve that returns the state to its initial value presses resolved distinctions back into an unread condition, and does so by the shape

of the spacetime rather than by any dynamical process acting on the traveller. That is the mechanism T28 names and forbids.

The conclusion of Part One can therefore be restated without the hedging it originally carried. It is not that the returned observer is arguably a different observer. It is that a model closing by erasure has, by T26, produced later states that are not records of the same system, and has therefore not returned anyone to their own past. It has produced a consistent loop containing a system that is not the one that set out.

T26 is PROVED in the Register; T28 is CONDITIONAL on A5, conservation of the readout. The application to the Deutsch and Gavassino conditions is PROPOSED, and it is the sharpest claim in this part.

30. What This Answers, and What It Owes

	Weak joint in Parts One to Three	What carries it	Grade
1	Order privileged over geometry without warrant	Alexandrov–Zeeman, Malament	PROVED, cited
2	Why the lift is the right repair	Gödel fails distinguishing; lift restores it	PROVED
3	What fixes the traversal index	T15, single monotone order	PROVED in Register
4	Why closure cannot touch the depth	D18 orthogonality of P_X and P_T	Definition
5	Why exactly one lift coordinate	Rank of P_T is one	PROPOSED
6	Whether erasure breaks identity	T26 and T28, overwrite forbidden	PROPOSED

What is owed is a single thing, and it is the same shape as the one Paper 21 Section Eleven owes at the representational end. Sections 27 to 29 identify Corpus objects with objects of the traversal argument: the clock mode with the traversal index, the closure mode with the lift coordinate, the record with the traveller's accumulated state. Each identification is natural and none is proved. If they fail, Sections 27, 28 and 29 revert together, and Sections 24 to 26 do not, because those rest on published theorems and on Gödel's causal structure rather than on the Corpus.

That is the right division of risk. The half of this part that answers the hardest objection is the half that does not depend on the framework at all.

31. Closing

Part One argued from the observer outward and reached a conclusion it could state but not secure: that a closed path repeats its geometry without reversing the order of its own traversal. The securing comes from two directions at once.

From outside the framework, the causal order is not subordinate to the geometry, because the geometry is reconstructed from it, and Gödel is precisely the case where the reconstruction fails because the order has been allowed to close. The lift is what restores the condition the reconstruction requires.

From inside it, the order of accumulation is single by theorem, orthogonal to the spatial directions by definition, retained as closure data rather than traversed, and protected from overwrite by a result about what it takes for later states to be records of the same system.

A closed path may repeat its geometry, but the geometry was recovered from the order in the first place. A theory that erases the order to close the path has spent the thing it was trying to preserve.

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